Local Probability Kernels, Weighted Ideal Counts, and Entropy Bookkeeping for Prime-Degree Torus Packets

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Abstract

We study the specific weighted ideal-sum interface in the September 24, 2026 manuscript *Equidistribution of Prime-Degree Torus Packets with Arbitrary Local Type*. Additive local Haar probability produces both an exact shell law and a finite Fourier contraction, with the index and removed Euler factors retained. A weighted Hölder argument proves an explicit local moment bound for every fixed $0 < s < 1/n$; at the source cutoff it gives every tail exponent below $1/(8n)$, improving the displayed $1/(32n)$. Conditional on the source's specified residue comparison and uniform multiplicative short-interval estimate, we extend its counting range to $x \geq Q^{\alpha}$, $\alpha > 1/2$, with relative widths down to $Q^{-\gamma}$ for $n\gamma < \alpha - 1/2$. We derive corresponding finite-scale ball bounds and improved cusp remainders. Finally, the cited entropy criterion yields the precise dimension-deficit constant and applies to every dominated invariant probability, hence almost every ergodic component. These are local proofs and conditional consequences, not an independent certification of the manuscript's global equidistribution claim. Contribution assessment: C3/5, explicit local moment and counting-range improvements with proved normalization and componentwise bookkeeping, while global arithmetic inputs and measure classification remain imported.

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1 Scope, status, and source architecture

The assignment is catalogue ID 34, applicability L3. Applicability measures the relevance of a Ram Calculus interface; C3 records the more limited contribution actually demonstrated here. Neither is a proof-validity or novelty certificate. The original seven TeX files were literally read in full, with bounded untruncated displays: 1,204 lines. The 123-line bibliography and 18-line README were also read. This is a complete literal read of this cached manuscript, not of its cited literature or the private corpus, and not a complete independent proof audit.

The source claims that volume-weighted packets for totally real fields of fixed prime degree $n \geq 5$, with arbitrary prescribed local homothety type, converge to Haar probability as multiplier-order discriminant tends to infinity. We preserve that as the *source claim*. The following map separates its arithmetic and dynamical dependencies.

Source file	Role and retained obligation
`paper.tex`	Main claim, field/order discriminants, prior work, sign-completed packet and scope; no individual-orbit conclusion.
`packets.tex`	Saturation, finite unit labels, orbit-volume measure, class-number normalization, exact ideal-sum unfolding.
`local.tex`	Shell mass, full-span marginal valuation bound, positive norm moment, large-supported-norm tail.
`analytic.tex`	Stark zero input, residue comparison, uniform Shiu input, removal of Euler factors, weighted ideal count.
`geometry.tex`	First moments for cusp cubes and vector boxes; tightness; fixed-radius Portmanteau argument.
`entropy.tex`	Tube covering, imported entropy criterion, restriction to collections of components, prime-degree classification.
`adelic.tex`	Compact torus quotient, arbitrary local module type, Haar pushforward and independence of auxiliary choices.

Labels in this report have the following meaning. A *local result* is proved here from stated elementary measure, linear algebra, or valuation facts. A *conditional consequence* uses named analytic or dynamical theorems as inputs, even if those inputs are established in the literature. An *imported assertion* has been located or read but its entire proof and all upstream hypotheses have not been independently reproved here. Finite tests concern manufactured models, not fields of unbounded discriminant.

2 Exact normalization dictionary

Fix a totally real field K of prime degree $n \geq 5$, an ordering $\sigma_1, \dots, \sigma_n$ of its real embeddings, its maximal order R, and a full $\mathbb{Z}$-lattice $M \subset K$. Put

$$J_0 = RM, \quad q = [J_0 : M], \quad D = |\operatorname{Disc} K|, \quad Q = q\sqrt{D}, \quad \kappa = \operatorname{Res}_{s=1} \zeta_K(s) > 0.$$

The multiplier order is $\mathcal{O}(M) = \{a \in K : aM \subset M\}$. Since $qR \subset \mathcal{O}(M) \subset R$,

$$D(M) = D[R : \mathcal{O}(M)]^2 \leq Dq^{2n} \leq Q^{2n}.$$

Thus $D(M) \to \infty$ forces $Q \to \infty$. The converse is not needed. Different local module types can have the same multiplier order.

The real space is $X_n = \operatorname{SL}_n(\mathbb{Z}) \backslash \operatorname{SL}_n(\mathbb{R})$ of covolume-one *row* lattices, with right action by

$$A_n = \{\operatorname{diag}(e^{u_1}, \dots, e^{u_n}) : \textstyle\sum u_i = 0\}.$$

Its Haar measure is $du_1 \cdots du_{n-1}$ with $u_n = -\sum_{i<n} u_i$, not Euclidean hypersurface measure. Include the source's distinct determinant-positive sign translates. The packet probability is the sum of orbit Haar measures divided by their total volume, not an unqualified equal-orbit average. If $C = \{t \in (\mathbb{R}^\times)^n : \prod |t_i| = 1\}$ has counting measure on its 2^n signs and deleted-coordinate logarithmic Haar measure, the source fundamental region F for $R^\times$ satisfies

$$\operatorname{vol} F = 2^{n-1} R_K, \qquad h_K \operatorname{vol} F = \sqrt{D}\, \kappa.$$

These are the source's regulator and class-number conventions; no extra $\sqrt{n}$, sign count, or local-label cardinality is inserted.

At a rational prime ℓ, normalize the saturated module to $L_\ell \subset R_\ell = R \otimes \mathbb{Z}_\ell$, with $R_\ell L_\ell = R_\ell$ and $[R_\ell : L_\ell] = q_\ell$. Let $S = \{\ell : q_\ell > 1\}$, so $q = \prod_S q_\ell$. For primes $\mathfrak{p} \mid \ell$, write $e_\mathfrak{p}, f_\mathfrak{p}$ for ramification and residue degrees; $\sum e_\mathfrak{p} f_\mathfrak{p} = n$. With $\mathbf{k} = (k_\mathfrak{p}) \geq 0$, set

$$d_\ell(\mathbf{k}) = \ell^{\sum f_\mathfrak{p} k_\mathfrak{p}}, \quad Z_\ell = \prod_{\mathfrak{p} \mid \ell} (1 - \ell^{-f_\mathfrak{p}})^{-1}, \quad Z_S = \prod_{\ell \in S} Z_\ell.$$

The source shell weight $P_\ell(\mathbf{k})$ is the probability, over Haar-uniform $z \in R_\ell^\times$, that a point with these valuations belongs to zL_ℓ. Equivalently it is the fraction of that additive valuation shell lying in L_ℓ. Units act transitively on each shell. Zero components have additive Haar measure zero. For a supported valuation tuple put $d = \prod_S d_\ell$ and $P(\mathbf{k}) = \prod_S P_\ell$. For an integral ideal $\mathfrak{a}$, $P(\mathfrak{a})$ uses precisely its valuations above S; it does not remove its coprime part.

2.1 Unfolding and the real Jacobian

For nonnegative Borel $f : \mathbb{R}^n \to [0, \infty]$, define $E_f(\Lambda) = \sum_{0 \neq v \in \Lambda} f(v)$ and

$$\mathcal{V}_f(z) = \sum_{\omega \in \{\pm 1\}^n} \int_{\mathbb{R}^{n-1}} f(\omega_1 e^{u_1}, \dots, \omega_{n-1} e^{u_{n-1}}, \omega_n z e^{-\sum_{i<n} u_i})\, \mathrm{d}u.$$

The exact source identity is

$$\int E_f \, \mathrm{d}\mu_{K,M} = \frac{1}{\sqrt{D}\, \kappa} \sum_{\mathfrak{a} \neq 0} P(\mathfrak{a}) \mathcal{V}_f(\mathrm{N}\mathfrak{a}/Q). \tag{1}$$

It is valid as an equality of extended nonnegative integrals, by Tonelli. The source proves it by uniform ideal-class and finite-label averaging, followed by unfolding free nonzero unit orbits. The common number of finite labels cancels; it is not q in general.

On each orthant the change from $(u_1, \dots, u_{n-1}, z)$ to x has absolute Jacobian one: factor x_i from each row and use determinant $1/z$, while $\prod |x_i| = z$. Consequently

$$\int_0^\infty \mathcal{V}_f(z)\,\mathrm{d}z = \int_{\mathbb{R}^n} f(x)\,\mathrm{d}x. \tag{2}$$

The fiber disintegration uses dz, whereas multiplicative convolution below uses dz/z. Confusing these measures shifts a Mellin line by one.

3 Local Fourier and probability representations

3.1 Finite Fourier normalization

Choose m with $\ell^m R_\ell \subset L_\ell$, and let $H = R_\ell/\ell^m R_\ell$, $\overline{L} = L_\ell/\ell^m R_\ell$. Give H normalized counting measure and its character group counting measure. For $g : H \to \mathbb{C}$ set

$$\widehat{g}(\chi) = |H|^{-1} \sum_{x \in H} g(x)\overline{\chi(x)}, \qquad \|g\|_H^2 = |H|^{-1} \sum_x |g(x)|^2.$$

Proposition 3.1 (Exact local projection and averaged contraction)**.** *Let $w(x) = \mathbb{E}_z \mathbf{1}_{z\overline{L}}(x)$, with uniform unit action induced from $R_\ell^\times$. Then*

$$\widehat{\mathbf{1}_{\overline{L}}}(\chi) = q_\ell^{-1} \mathbf{1}_{\overline{L}^\perp}(\chi), \qquad \|\mathbf{1}_{\overline{L}}\|_H^2 = q_\ell^{-1}.$$

The normalized convolution $Tg = (q_\ell w) * g$ is a positive self-adjoint contraction on $L^2(H)$ with multipliers

$$m_\chi = \mathbb{E}_z \mathbf{1}_{(z\overline{L})^\perp}(\chi) \in [0,1], \qquad \|Tg\|_H^2 = \sum_\chi m_\chi^2 |\widehat{g}(\chi)|^2.$$

It preserves constants. It need not be an orthogonal projection.

Proof. A character sum over a subgroup is its cardinality if the character is trivial there, and zero otherwise, by translation by an element on which the character is nontrivial. Since $|\overline{L}|/|H| = q_\ell^{-1}$, the first formula follows. Character orthogonality proves Plancherel. Average the subgroup formula over units; Fourier transform turns normalized convolution into multiplication. Real nonnegative multipliers prove the assertions. Averaging several different annihilators can give intermediate eigenvalues, so idempotence is not automatic. No independence of valuation coordinates is used. □

This is an exact finite-dimensional energy interface, not a spectral gap for the arithmetic packet. At growing m, near-zero frequencies and varying L_ℓ still require separate estimates.

3.2 Trace-dual Haar convention

For $K_\ell = K \otimes \mathbb{Q}_\ell$, use a character of $\mathbb{Q}_\ell$ with conductor $\mathbb{Z}_\ell$ and the trace pairing. Normalize additive dx by $\mathrm{vol}(R_\ell) = 1$. The annihilator of R_ℓ is the inverse different $\mathfrak{D}_\ell^{-1}$, not generally R_ℓ. If $D_\ell = [\mathfrak{D}_\ell^{-1} : R_\ell] = \ell^{v_\ell(D)}$, then the dual Plancherel measure under this identification is $dy = D_\ell^{-1} dx$. Indeed the transform of $\mathbf{1}_{R_\ell}$ is $\mathbf{1}_{\mathfrak{D}_\ell^{-1}}$, and its two squared norms must both be one. Character orthogonality on compact open subgroups gives more generally

$$\widehat{\mathbf{1}_{L_\ell}}(y) = q_\ell^{-1} \mathbf{1}_{L_\ell^\perp}(y), \qquad \mathrm{vol}_{dy}(L_\ell^\perp) = q_\ell.$$

Self-dual Haar instead assigns R_ℓ volume $D_\ell^{-1/2}$. The finite quotient formulation avoids choosing a trace-dual identification, but never licenses dropping the different in a local field Fourier calculation.

3.3 The exact shell law

Proposition 3.2 (Probability normalization)**.** *For additive Haar probability on L_ℓ, its valuation tuple has law*

$$\pi_\ell(\mathbf{k}) = \frac{q_\ell}{Z_\ell}\frac{P_\ell(\mathbf{k})}{d_\ell(\mathbf{k})}.$$

Thus $\pi = \prod_{\ell\in S}\pi_\ell$ is a probability and

$$\sum_{\mathbf{k}} \frac{P(\mathbf{k})}{d} = \frac{Z_S}{q}, \qquad \sum_{\mathbf{k}} P(\mathbf{k})d^{s-1} = \frac{Z_S}{q}\mathbb{E}_\pi d^s \tag{3}$$

whenever the latter nonnegative moment is defined.

Proof. The ambient shell has additive mass $Z_\ell^{-1}d_\ell^{-1}$, and fraction P_ℓ belongs to L_ℓ. Divide its intersection mass by $\mathrm{vol}(L_\ell) = q_\ell^{-1}$, and multiply over rational primes. Independence here is across separately sampled local modules, not across prime-ideal coordinates inside one module. □

4 A sharper local moment and tail

Theorem 4.1 (Weighted marginal bound)**.** *For every integer $n \geq 2$, every degree-n finite product of local extensions of $\mathbb{Q}_\ell$, and every full $\mathbb{Z}_\ell$-module $L \subset R_\ell$ with $R_\ell L = R_\ell$, additive Haar probability on L satisfies, for every fixed $0 < s < 1/n$,*

$$\mathbb{E}_L d_\ell^s \leq C_\ell(n,s) := \frac{1-\ell^{-1}}{1-\ell^{ns-1}}. \tag{4}$$

The bound is independent of the index of L, its type, and the ramification pattern.

Proof. Full R_ℓ-span means the projection of L to each local factor contains a unit. Modulo $\mathfrak{p}^k$, the image therefore contains the scalar multiples of a unit by $\mathbb{Z}_\ell$, a group of cardinality $\ell^{\lceil k/e_\mathfrak{p}\rceil}$. Haar probability pushes forward uniformly to this finite image, so

$$\mathbb{P}_L(v_\mathfrak{p} \geq k) \leq \ell^{-\lceil k/e_\mathfrak{p}\rceil}.$$

Put $J_\mathfrak{p} = \lceil v_\mathfrak{p}/e_\mathfrak{p}\rceil$. For each integer $a \geq 1$, the event $J_\mathfrak{p} \geq a$ is $v_\mathfrak{p} \geq e_\mathfrak{p}(a-1)+1$, whence its probability is at most ℓ^{-a}. With $t = \ell^{ns} < \ell$, summation of nonnegative tails gives

$$\mathbb{E}t^{J_\mathfrak{p}} = 1 + (t-1)\sum_{a\geq 1} t^{a-1}\mathbb{P}(J_\mathfrak{p} \geq a) \leq 1 + \frac{t-1}{\ell - t} = C_\ell(n,s). \tag{5}$$

Let $\alpha_\mathfrak{p} = e_\mathfrak{p}f_\mathfrak{p}/n$. These positive weights sum to one, and $d_\ell^s \leq \prod_\mathfrak{p}(\ell^{nsJ_\mathfrak{p}})^{\alpha_\mathfrak{p}}$. Weighted Hölder, with no independence assumption, bounds the expectation by $\prod(\mathbb{E}\ell^{nsJ_\mathfrak{p}})^{\alpha_\mathfrak{p}} \leq C_\ell$. This also covers a single local factor, for which the Hölder step is just equality of its single weight with one. □

Corollary 4.2 (Variable supported-norm cutoff)**.** *For fixed n, $0 < s < 1/n$, $\varepsilon > 0$, and every q,*

$$\sum_{\mathbf{k}} P(\mathbf{k})d^{s-1} \leq \frac{1}{q}\prod_{\ell\mid q} Z_\ell C_\ell(n,s) \ll_{n,s,\varepsilon} q^{-1+\varepsilon}. \tag{6}$$

If $q \le Q$ and $\rho > 0$, then

$$\sum_{d > Q^\rho} \frac{P(\mathbf{k})}{d} \ll_{n,s,\varepsilon} q^{-1} Q^{-\rho s + \varepsilon}. \tag{7}$$

For every fixed $0 < \delta < \rho/n$ this is $O_{n,\rho,\delta}(q^{-1}Q^{-\delta})$.

Proof. Use Proposition 3.2 and Theorem 4.1. For fixed $s < 1/n$, $Z_\ell C_\ell = 1 + O_{n,s}(\ell^{-(1-ns)} + \ell^{-1})$. For all sufficiently large primes this is at most ℓ^ε; the finitely many smaller prime factors contribute a constant, since each occurs only once in the product. Their product is at most a constant times q^ε. Markov's inequality on the nonnegative sum proves (7). Choose $\delta/\rho < s < 1/n$ and $0 < \varepsilon < \rho s - \delta$. □

At $\rho = 1/8$, every $\delta < 1/(8n)$ is available, in place of the source's fixed $1/(32n)$. Constants deteriorate as $s \uparrow 1/n$; there is no uniform endpoint assertion. The source's union-bound route also extends to variable s, with local factor $1 + n(1-\ell^{-ns})\ell^{ns-1}/(1-\ell^{ns-1})$. The weighted Hölder proof removes this extra factor n in the tail increment.

4.1 What fails, and what is not ruled out, at the endpoint

For the split algebra $R_\ell = \mathbb{Z}_\ell^n$, consider the genuine full-rank module

$$L_H = \{(x_1, \dots, x_n) : x_i \equiv x_1 \pmod{\ell^H}\}, \qquad H \ge 1.$$

It contains the all-ones vector, has full R_ℓ-span, and index $q_\ell = \ell^{H(n-1)}$. For Haar-uniform $x \in L_H$, x_1 is uniform in $\mathbb{Z}_\ell$. On $v_\ell(x_1) = k < H$, all coordinates have valuation k. Hence

$$\mathbb{E}_{L_H} d_\ell^s \ge (1 - \ell^{-1}) \sum_{k=0}^{H-1} \ell^{(ns-1)k}. \tag{8}$$

At $s = 1/n$ this grows like H, disproving an index-independent constant bound. It *does not* disprove a bound with q_ℓ^ε: H is logarithmic in the index. For $s > 1/n$ it grows as a positive power of q_ℓ and rules out such a bound for all sufficiently small ε. For $s < 1/n$, the lower bound tends to $C_\ell(n,s)$; thus (4) is the exact supremum over these local types. We make no global realization or endpoint impossibility claim beyond these statements.

5 Ideal-sum kernels and energy residuals

Write $b(m) = a_K(m)\mathbf{1}_{(m,q)=1}$, where $a_K(m)$ counts integral ideals of norm m. For a compactly supported bounded function $\phi : (0,\infty) \to \mathbb{C}$ define

$$F_\phi(t) = \frac{Z_S}{\kappa t} \sum_{m \ge 1} b(m)\phi(m/t), \qquad W_\phi(t) = \frac{q}{\kappa t} \sum_{\mathfrak{a} \ne 0} P(\mathfrak{a})\phi(\mathrm{N}\mathfrak{a}/t).$$

These sums are finite for each $t > 0$. Unique ideal factorization and (3) give the exact identity

$$W_\phi(t) = \mathbb{E}_\pi F_\phi(t/d). \tag{9}$$

Indeed, after splitting off the S-supported part, its coefficient is $qP/(Z_S d) = \pi$. This is where the removed Euler factor cancels; omitting Z_S from F_ϕ changes the normalization.

Let ν be the law of $\log d$ under π, a probability on $[0,\infty)$. Then (9) is convolution by ν in $v=\log t$. For every $G\in L^2(\mathbb{R},dv)$,

$$\|\nu * G\|_2 \le \|G\|_2, \qquad \widehat{\nu * G}(\xi)=\widehat{\nu}(\xi)\widehat{G}(\xi), \quad \widehat{\nu}(\xi)=\mathbb{E}_\pi e^{-i\xi\log d}, \quad |\widehat{\nu}|\le 1. \tag{10}$$

For our Fourier convention $\widehat{G}(\xi)=\int G(v)e^{-i\xi v}dv$, Plancherel has factor $1/(2\pi)$. Jensen and translation invariance prove the contraction; finite measures and L^2 approximation prove the multiplier identity. A raw arithmetic profile $F_\phi(e^v)$ need not belong to $L^2(\mathbb{R})$. Apply this assertion to an explicitly square-integrable residual or to a supplied windowed profile, not automatically to the unwindowed count. Windowing before convolution changes the boundary terms and must be recorded.

For $\nu_{\le T}=\nu|_{[0,\log T]}$, positivity and Minkowski give

$$\|\nu * G-\nu_{\le T} * G\|_2 \le \pi(d>T)\|G\|_2.$$

The truncated kernel is a subprobability, not a renormalized probability. Moreover $\pi(d>T)\le T^{-s}\prod_{\ell\mid q}C_\ell(n,s)$. If a residual is split orthogonally as $G=G_0+R$, then $\|\nu * R\|_2\le\|R\|_2$, but the two images need not remain orthogonal; the output energy includes their cross term. A unitary or Parseval coordinate change preserves this energy and cannot supply missing arithmetic cancellation.

These restrictions follow the actual private proof routes read here: the master-atoms Mellin inversion/energy proof uses $g_c(v)=e^{cv}f(e^v)$ and energy $\int|f(t)|^2t^{2c-1}dt$; its $c=0$ line is precisely dt/t. TR1674's finite Gram and carry/cross/noise identities retain the arithmetic input and its boundary terms. We transfer that discipline, not any RH-related conclusion or an assertion that local positivity forces global equidistribution.

6 A quantified extension of the weighted count

The following two inputs are *imported*, with the same totally real fixed-prime-degree scope as the source.

R: residue comparison. $h\zeta_K(1+h)\asymp_n\kappa$ uniformly for $0<h\le 4/\log D$, and $\kappa^{-1}\ll_n\log(3D)$. The ratio is not asserted to tend uniformly to one. The source's log-derivative integration allows factors $\exp(O_n(h\log D))$. Its bounded-discriminant completion uses fixed prime degree.

S: uniform Shiu bound. For nonnegative multiplicative $b\le d_n$ and $u^{1/2}\le v\le u$,

$$\sum_{u-v<m\le u} b(m)\ll_n \frac{v}{\log u}\exp\Big(\sum_{\ell\le u}\frac{b(\ell)}{\ell}\Big)$$

for sufficiently large u, uniformly in the function b in this class. The source obtains this from the corrected Pollack theorem with empty excluded residue sets and fixed exponent $1/4$; its prime-power and subpower constants depend only on n.

Theorem 6.1 (Conditional shorter relative intervals)**.** *Assume R and S. Fix $n\ge 5$ prime, $A_0\ge 1$, $1/2<\alpha<1$, and $0\le\gamma<(\alpha-1/2)/n$. Uniformly for the source's fields and full local types, for all sufficiently large Q depending only on these fixed parameters,*

$$\sum_{x-y<\mathrm{N}\mathfrak{a}\le x} P(\mathfrak{a})\ll_{n,\alpha,\gamma}\frac{\kappa y}{q}, \quad Q^\alpha\le x\le A_0Q, \quad xQ^{-\gamma}\le y\le x. \tag{11}$$

For every fixed $\eta>0$, the same conclusion holds for $\eta x\le y\le x$ with threshold also depending on η.

Proof. Choose constants

$$n\gamma < \rho < \alpha - \tfrac{1}{2}, \qquad \gamma/\rho < s < 1/n, \qquad 0 < \varepsilon < \rho s - \gamma.$$

For $\gamma = 0$ these choices still exist. Splitting ideals into S-supported and coprime parts expresses the left side as

$$\sum_{\mathbf{k}} P(\mathbf{k}) \sum_{(x-y)/d < m \le x/d} b(m).$$

For $d \le Q^{\rho}$, put $u = x/d$, $v = y/d$. Then $u \ge Q^{\alpha-\rho} > Q^{1/2} \ge D^{1/4}$, so $h = 1/\log u$ is within R. Also $2\gamma < \alpha - \rho$: indeed $\gamma < 1/(2n) \le 1/10$, whereas $\alpha - \rho > 1/2$. Thus $v \ge uQ^{-\gamma} \ge \sqrt{u}$ for large Q.

For completeness the Euler-factor step retains all its arithmetic data. Using $a_K(\ell) \le n$,

$$\sum_{\ell \le u} \frac{a_K(\ell)}{\ell} \le \log \zeta_K(1 + 1/\log u) + O_n(1) \le \log(\kappa \log u) + O_n(1).$$

The first error follows from $1 - \ell^{-h} \le h \log \ell$ and $\sum_{\ell \le u} \log \ell/\ell \ll \log u$. The exact Euler factors satisfy $\log Z_S = \sum_{\ell \mid q} a_K(\ell)/\ell + O_n(1)$, including ramified primes. The omitted primes $\ell \mid q$, $\ell > u$, contribute at most $n \log(2q)/(u \log 2) = o(1)$ uniformly. Consequently

$$\exp\Bigl(\sum_{\ell \le u} \frac{b(\ell)}{\ell}\Bigr) \ll_n \frac{\kappa \log u}{Z_S}.$$

Input S and (3) bound the small-d sum by $C_n \kappa y Z_S^{-1} \sum P/d = C_n \kappa y/q$.

For $d > Q^{\rho}$, terms with $d > x$ vanish. Elementary divisor summation gives $\sum_{m \le u} d_n(m) \le u(1 + \log u)^{n-1}$ for $u \ge 1$; one can prove this by induction using $d_n = 1 * d_{n-1}$. Corollary 4.2 therefore bounds the large-d sum by

$$O_{n,s,\varepsilon}\left(\frac{x}{q} Q^{-\rho s + \varepsilon} (1 + \log x)^{n-1}\right).$$

Relative to $\kappa y/q$, this is $O(Q^{\gamma - \rho s + \varepsilon}(\log(3Q))^n) = o(1)$, using $D \le Q^2$ and fixed A_0. This proves (11). For fixed η the same proof takes $v \ge \eta u$, and replaces the relative factor Q^{γ} by η^{-1}. The threshold absorbs it. All parameter choices are fixed before taking $Q \to \infty$. □

The source uses $\alpha = 3/4$, cutoff $\rho = 1/8$, fixed η and one moment. The new range permits every fixed $\alpha > 1/2$ and a specified shrinking relative width. This is an upper bound, not an asymptotic with leading constant one. It does not prove R or S anew, reach $\alpha = 1/2$, or yield uniformity as γ approaches its boundary.

7 Nonescape and finite-scale vector boxes

Corollary 7.1 (Conditional cusp remainder)**.** *Under the unfolding identity and inputs R and S, fix $1/2 < \alpha < 1$. Uniformly for $0 < \theta < 1$ and sufficiently large Q,*

$$\mu_{K,M}\{\Lambda : \exists 0 \ne v \in \Lambda, \|v\|_{\infty} < \theta\} \ll_{n,\alpha} \theta^n + Q^{\alpha-1}(\log Q)^{n-1}.$$

In particular one can take a remainder $Q^{-1/2+\delta}(\log Q)^{n-1}$ for every fixed $0 < \delta < 1/2$.

Proof. For the cube test the source's fiber integral is $2^n(\log(\theta^n/z))_+^{n-1}/(n-1)!$. Put $X=\theta^nQ$, $U=Q^\alpha$. If $X<1$ the ideal sum is empty. The terms up to $\min(X,U)$ are bounded using Theorem 6.1 at $x=y=U$ and the upper logarithmic weight $(\log Q)^{n-1}$. After division by $\sqrt{D}\kappa$ their contribution is $O((U/Q)(\log Q)^{n-1})$. For the remaining terms use $A_U(t)=\sum_{U<\mathrm{N}\mathfrak{a}\le t}P(\mathfrak{a})\ll\kappa t/q$. Tonelli gives

$$(n-1)\int_U^X A_U(t)(\log(X/t))^{n-2}\frac{dt}{t}\ll\kappa X/q.$$

There is no lower boundary term because $A_U(U)=0$, and the upper weight vanishes. Divide by $\sqrt{D}\kappa$, use $Q=q\sqrt{D}$, and bound the cusp indicator by the nonzero-vector count. □

Corollary 7.2 (Conditional polynomial radius range)**.** *Assume R, S and the source unfolding. For every fixed $0<\gamma<1/(2n)$ and compact $\Omega\subset X_n$, there are $C_\Omega,r_\Omega,Q_\Omega>0$ such that*

$$\mu_{K,M}(xB(r))\le C_\Omega r^n\quad(x\in\Omega,\ Q\ge Q_\Omega,Q^{-\gamma}\le r\le r_\Omega),$$

where $B(r)=\{h\in\mathrm{SL}_n(\mathbb{R}):\max_{ij}|h_{ij}-\delta_{ij}|<r\}$. Constants also depend on n,γ.

Proof. As in the source, a finite compact cover supplies at each $x\in\Omega$ a lattice vector w_x in one compact $V\subset(\mathbb{R}^\times)^n$. To justify existence, for a basis use integer coefficient vectors $(1,N,\dots,N^{n-1})$ and avoid the finitely many roots of the nonzero coordinate polynomials. Continue each chosen vector in a local chart and take a finite cover. No measurable selection is needed. Every lattice in $xB(r)$ contains a vector in $w_x+[-b_0r,b_0r]^n$ for one uniform b_0. The fiber of this box has height $O(r^{n-1})$ and product support within $O(r)$ of $z_x=\prod|(w_x)_i|$, bounded above and below on V.

Choose $\alpha<1$ with $n\gamma<\alpha-1/2$, then $\gamma<\gamma_1<(\alpha-1/2)/n$. A slightly enlarged supported-norm interval has endpoint $x'\asymp_V Q$ and width $y'\asymp_V rQ$, with $y'\le x'$ for r_Ω small. Uniformly in the stated radius range, $y'/x'\ge c_VQ^{-\gamma}\ge Q^{-\gamma_1}$ eventually, and $x'\ge Q^\alpha$. Theorem 6.1 gives a first moment $O(r^{n-1}\kappa rQ/(q\sqrt{D}\kappa))=O(r^n)$. The vector count majorizes the ball indicator, proving the claim. □

This upgrades the source's fixed-r threshold to a proved polynomial range, conditional on the same analytic inputs. It is not an unrestricted microscopic estimate: at still smaller radii the packet is supported on lower-dimensional orbits. For weak limits one first fixes $r>0$, then lets $Q\to\infty$ and applies Portmanteau to the open ball. Mahler's compactness criterion plus Corollary 7.1 supplies tightness by choosing θ first. A probability limit, rather than an unnormalized surviving measure, is essential in the entropy step.

8 Dimension deficit and every dominated component

Here the local geometric proof is independent of the arithmetic count. The noncompact entropy criterion of ELMV [2, Corollary 3.3] is imported: flow-invariant probabilities converging to a probability, with tube masses $O_\Omega(e^{-2\eta t_i})$ uniformly at centers in each compact set and $t_i\to\infty$, have limit entropy at least η. We inspected this statement and its Fubini reduction, not the complete proof of its antecedent Proposition 3.2.

Theorem 8.1 (Conditional dimension-deficit entropy)**.** *Let $n\ge3$ and μ be an A_n-invariant probability on X_n. Suppose for some $\sigma>n-1$ and $B\ge0$, on every compact Ω,*

$$\mu(xB(r))\le C_\Omega r^\sigma(1+|\log r|)^B\quad(x\in\Omega,\ 0<r<r_\Omega).$$

Fix distinct $\lambda_i \in \mathbb{R}$ summing to zero and put $a(t) = \operatorname{diag}(e^{\lambda_i t})$, $\lambda = \min_{i \neq j} |\lambda_i - \lambda_j| > 0$. Under the cited criterion, every A_n-invariant probability $\xi \leq c\mu$ with $c < \infty$ satisfies

$$h_\xi(a(1)) \geq \frac{\lambda}{2}(\sigma - n + 1). \tag{12}$$

The same bound holds for almost every A_n-ergodic component of μ, using the entropy integral formula over the invariant component factor.

Proof. Fix $0 < b < 1/4$ and $T_t = a(t)B(b)a(-t) \cap a(-t)B(b)a(t)$. Its off-diagonal entries are bounded by $be^{-|\lambda_i - \lambda_j|t}$ and its diagonal entries stay near one. Set $r = e^{-\lambda t}$. Every non-diagonal determinant term contains at least two off-diagonal entries, so the diagonal product is $1 + O(r^2)$. Replacing its last entry by the reciprocal product of the others gives a diagonal matrix $d(h)$ in one compact subset $\mathcal{D} \subset A_n$, with $d(h)^{-1}h \in B(Cr)$. A logarithmic mesh covers $\mathcal{D}$ by $O(r^{-(n-1)})$ balls. Multiplication changes constants but not this exponent. Thus, for x in a compact Ω, all centers lie in $\Omega\mathcal{D}$ and

$$\xi(xT_t) \ll_{\Omega,c} e^{-\lambda(\sigma-n+1)t}(1+t)^B.$$

Use the constant sequence $\xi_i = \xi$ and $t_i = i$ in the imported criterion. Every $\eta < \lambda(\sigma - n + 1)/2$ is allowed, because an exponential with a strict exponent gap absorbs the polynomial. Taking the supremum over such η proves (12). For $B = 0$ one may directly take equality; a larger constant handles initial times.

Write $\mu = \int \nu_z \, \mathrm{d}\tau(z)$ for its A_n-ergodic decomposition. The entropy function is measurable via a countable generating sequence of finite partitions and the usual infimum of normalized joined-partition entropies. If the set where $h_{\nu_z}(a(1)) < H$ has positive measure, where $H = \lambda(\sigma - n + 1)/2$, one of the sets $Z_j = \{h_{\nu_z}(a(1)) \leq H - 1/j\}$ has positive weight w. Its normalized mixture $\xi = w^{-1} \int_{Z_j} \nu_z \, \mathrm{d}\tau$ satisfies $\xi \leq w^{-1}\mu$. The entropy integral formula applies because the component factor is fixed by $a(1)$; it does not require the ν_z to be individually $a(1)$-ergodic. It gives $h_\xi(a(1)) \leq H - 1/j$, contradicting (12). □

For the source ball exponent $\sigma = n$ this yields $\lambda/2$, rather than the stated $\lambda/3$. This is sharper use of the cited criterion, not a new entropy theorem. It also shows that a loss of any fixed amount less than one in the ball exponent, and fixed logarithmic losses, leave a positive entropy conclusion. Individual ergodic components need not satisfy $\nu_z \leq c\mu$; the restricted mixture is the indispensable step. Positive entropy of μ alone is insufficient: a mixture can contain a positive-weight zero-entropy component.

Only after this step can one invoke EKL [4, Theorem 1.3 and Corollary 1.4]: for prime n, the relevant A_n-ergodic positive-entropy probabilities are Haar. The inversion map $\Gamma g \mapsto g^{-1}\Gamma$ changes our right action to left action by $a(-t)$; entropy is unchanged by inversion of an invertible map. We do not extend that classification to composite degree, noninvariant measures, or subprobability limits.

9 Public context, comparison, and limitations

The October 8, 2026 lookup used only public titles and primary public pages. ELMV's cubic paper combines subconvexity, measure classification and local harmonic analysis; it is context, not a proof of this arbitrary-prime-degree claim [3]. Lemke Oliver–Thorner–Zaman give an almost-all prime-degree application, a different quantifier from every packet in the source [7]. The primary EKL statement explicitly includes the prime-degree specialization. The source's Stark input is not restricted to normal fields: Lemma 8 uses constant 4 in the normal case and $4n!$ otherwise, and

forces a quadratic subfield from a sufficiently near-one real zero [5, p. 145]. Odd degree excludes that subfield. We visually read that page and its lemma proof, not Stark's complete proof chain. This removes a possible misreading of normality, but does not certify every analytic step of the packet manuscript.

Advantages. The probability law isolates exactly the Z_S/q mass; the finite Fourier description fixes both counting and dual measures. Hölder handles correlated local coordinates and gives a locally sharp index-independent subcritical moment. The cutoff proof makes parameter slack explicit and supplies a genuine conditional counting-range extension. The geometry and entropy steps record the order of limits, the dimension threshold, and the domination needed on restrictions.

Costs and limitations. There is no free improvement in the arithmetic input: uniform R and S, ideal coefficients at ramified primes, and the exact unfolding all remain essential. A contraction in logarithmic L^2 gives no pointwise short-interval bound without additional hypotheses. Endpoint constants can diverge, and the q^ε endpoint remains undecided here. The argument yields upper bounds rather than a power-saving equidistribution rate or a full asymptotic. Entropy and classification are substantial imported results, not consequences of a Gram matrix. No independent global certification or priority claim is made.

Feasible further work. First, investigate $s = 1/n$ with index-dependent logarithmic factors; the diagonal congruence family supplies a required lower benchmark, not a disproof of q^ε bounds. Second, expose explicit constants and discriminant thresholds in R and S before numerical field-level validation. Third, compare the conditional radius exponent with actual packet spacing in enumerated local types. Fourth, study anisotropic tube coverings using the individual root gaps rather than their minimum, while preserving domination on every restricted component mixture. None of these tasks is asserted complete.

10 Evidence and verification boundaries

The literal ledger contains all nine cached public text files and the following actual private intervals, whose proofs were read locally: master-atoms chapter, lines 447–543 (Mellin substitution, inversion, Plancherel and convolution); TR1674 revised version, lines 80–399 (definitions, exact convolution and energy proofs, quarter-scale terms, finite Gram/Parseval facts, and stated analytic boundaries). These are 417 private lines in two files, not complete private-file or whole-corpus coverage. Remaining private lines and all other private manuscripts are unread for this report. No private title or excerpt was sent to a public search service.

Useful public-source proof locators are: `local.tex` 32–97 for shells and marginal tails; `analytic.tex` 1–102 for residue and Shiu inputs and 103–189 for the weighted count; `geometry.tex` 18–73 for cusp integration and 92–150 for vector boxes and limits; `entropy.tex` 30–88 for tube geometry and the cited criterion, 96–136 for component restriction, and 138–167 for classification and the claimed conclusion. The source normalization and unfolding are in `packets.tex`; its entire 215 lines were read. All seven TeX files' architecture was mapped before formulation. Full literal reading alone does not independently establish every external theorem or every arithmetic uniformity assertion.

The author-owned reproducible tests enumerate finite local modules, character normalizations and correlated shells; check tail identities, cutoff feasibility and kernel energy; and verify explicit endpoint benchmarks. These are bounded exact-rational or declared floating-point sanity tests, not

asymptotic arithmetic experiments. Parent checks are not counted as author tests. The separate test JSON gives exact counts, tolerances, and failures. Native compilation is attempted before the unchanged MiKTeX export helper. The visual ledger records inspection of every final rendered page. The read-scope, proof, identity, literature, test, visual, assessment and freeze records preserve hashes and scope.

Parent provenance identifies unchanged cached source text between these two snapshots:

```
Pinned:  adc7f1241b42e322a6451854ab7e4b4c146bf78a
Oct. 8:  fd4aeeb2ee4fc729c18d98444fed42fd0529eeeb
```

Our hashes independently check those nine files. This is snapshot provenance, not live-branch status, publisher-based validity, or an independent proof audit. Mutable parent files are not frozen as author evidence.

References

[1] OpenAI, *Equidistribution of Prime-Degree Torus Packets with Arbitrary Local Type*, September 24, 2026. Cached source snapshot specified in Section 10. Public manuscript directory.

[2] M. Einsiedler, E. Lindenstrauss, P. Michel and A. Venkatesh, *Distribution of periodic torus orbits on homogeneous spaces*, Duke Math. J. 148 (2009), 119–174, especially Corollary 3.3. Author-hosted text; DOI.

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[4] M. Einsiedler, A. Katok and E. Lindenstrauss, *Invariant measures and the set of exceptions to Littlewood's conjecture*, Ann. of Math. 164 (2006), 513–560. Author-hosted text.

[5] H. M. Stark, *Some effective cases of the Brauer–Siegel theorem*, Invent. Math. 23 (1974), 135–152, Lemma 8 on p. 145. Primary journal scan.

[6] P. Pollack, *Nonnegative multiplicative functions on sifted sets, and the square roots of* -1 *modulo shifted primes*, Glasgow Math. J. 62 (2020), 187–199. Corrected author version, Theorem 1.1.

[7] R. J. Lemke Oliver, J. Thorner and A. Zaman, *An approximate form of Artin's holomorphy conjecture and non-vanishing of Artin L-functions*, Invent. Math. 235 (2024), 893–971. Author manuscript record.

[8] M. Subbarao, private Ram Calculus volume 2, master-atoms chapter, selected Mellin proof interval specified in Section 10. Local only; no public URL.

[9] M. Subbarao, private TR1674 revised version, selected energy and Gram proof intervals specified in Section 10. Local only; no public URL.